$Fibonacci\ Series\ -\ Growth\ Rate\ of\ Output$

vs

$Time\ Complexity$

$Here\ we\ have$:

```
    return fibonacci(n - 1) + fibonacci(n - 2);
```

$Time\ Complexity: T(n) = T(n-1) + T(n-2) + O(1)$

$Growth\ Rate\ of\ Output: F(n) = F(n-1) + F(n-2)$

$1)\ First\ Method\ \big(Assumming\ F(n-1) \approx F(n-2)\big)\ \begin{bmatrix} Not \\ Precise \end{bmatrix}$

$From\ time\ complexity\ we\ came\ to\ know\ that$:

$$T(n) = \begin{cases} 1 & if(n = 0) \\ 1 & if(n = 1) \\ T(n-1) + T(n-2) & if(n > 1) \end{cases}$$

$F(n) = F(n-1) + F(n-2)$

$Assumming\ F(n-1) \approx F(n-2)\ Then$:

$F(n) = F(n-1) + F(n-1)$

$F(n) = 2F(n-1)$

$Hence\ , for\ F(n-1) = 2 \times 2F(n-2) = 4F(n-2) = 2^2F(n-2)$

$Therefore\ , substituting\ , we\ get$:

$F(n) = 2^2F(n-2)$

$By\ doing\ this\ upto\ `i`\ times\ we\ get$:

$F(n) = 2^iF(n-i)$

$If\ , n-i = 1\ , then : i = n-1\ , replacing\ `i`\ , we\ get$:

$F(n) = 2^{n-1}F\big(n-(n-1)\big)$

$F(n) = 2^{n-1}F(1)$

$F(n) = 2^{n-1} \times 1\ [As, F(1) or\ T(1) = 1]$

$F(n) = 2^{n-1}$

$We\ can\ also\ write\ it\ as$:

$F(n) = 2^n \times 2^{-1}$

$F(n) = 2^n \times \frac{1}{2}$

$Applying\ theta$ `Θ` we get:

$F(n) = \Theta(2^n) \times \Theta\left(\frac{1}{2}\right)$

$F(n) = \Theta(2^n)\ [This\ is\ not\ precise\ .]$

$2)$ $\boldsymbol{Assuming\ F(n) = x^n}$

$\boldsymbol{Why\ this\ guess}$?
$\boldsymbol{Because}$:

- $\boldsymbol{Recurrences\ often\ behave\ like\ exponentials}$
- $\boldsymbol{Similar\ to\ how\ linear\ differential\ equations\ use\ e^x}$

$\boldsymbol{Therefore}$, $\boldsymbol{we\ replace\ every\ term\ as}$:

$$\boldsymbol{F(n) = F(n-1) + F(n-2)}$$

$$\boldsymbol{x^n = x^{n-1} + x^{n-2}}$$

we want to *eliminate the dependence on* `$\boldsymbol{n}$` ***and get a*** *constant equation in* $\boldsymbol{x}$.

We want to convert everything into the ***same power base level***.

So we divide by the ***smallest power present***, *which is*: x^{n-2}

$\boldsymbol{Dividing\ both\ side\ with\ `x^{n-2}`\ we\ get}$:

$$\boldsymbol{\frac{x^n}{x^{n-2}} = \frac{x^{n-1}}{x^{n-2}} + \frac{x^{n-2}}{x^{n-2}}}$$

$$\boldsymbol{\Longrightarrow x^{n-(n-2)} = x^{n-1-(n-2)} + x^{n-2-(n-2)}}$$

$$\boldsymbol{\Longrightarrow x^2 = x^1 + x^0}$$

$$\boldsymbol{\Longrightarrow x^2 = x + 1}$$

$$\boldsymbol{\Longrightarrow x^2 - x - 1 = 0}$$

$\boldsymbol{By\ Quadratic\ Formula}$:

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\boldsymbol{Where, a = 1\ , b = -1\ and\ c = -1}$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 1 \times -1}}{2 \times 1}$$

$$= \frac{1 \pm \sqrt{1 + 4}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

$\boldsymbol{So\ we\ get}$:

$$\varphi = \frac{1 + \sqrt{5}}{2} = \frac{1 + 2.236}{2} = \frac{3.236}{2} \approx 1.618$$

$$\psi = \frac{1 - \sqrt{5}}{2} = \frac{1 - 2.236}{2} = \frac{-1.236}{2} \approx -0.618$$

$$F(n) = A\varphi^n + B\psi^n$$

$\boldsymbol{As, \psi^n =}$

$\boldsymbol{(-0.618)^2 = 0.382}$

$\boldsymbol{(-0.618)^3 = -0.236}$

$\boldsymbol{(-0.618)^4 = 0.145}$

$\boldsymbol{(-0.618)^5 = -0.090}$

……

$\boldsymbol{Hence\ we\ can\ tell:\ \psi^n \to 0}$.

$\boldsymbol{Where, \varphi^n = (1.618)^n \to \infty}$

Term	*Value*	*Behavior*
$\boldsymbol{\varphi}$	1.618	*grows exponentially*
$\boldsymbol{\psi}$	-0.618	*shrinks to* 0

$\boldsymbol{Therefore\ we\ can\ say: \Theta(\varphi^n), where\ `\varphi = 1.618`}$.

$\boldsymbol{And, its\ precise\ and\ Actual.}$

$\boldsymbol{Also, in\ Time\ Complexity\ we\ have\ seen}$:

$\boldsymbol{T(n) = T(n-1) + T(n-2) + O(1)\ or, T(n-1) + T(n-2) + 1}$
$\boldsymbol{\Rightarrow O(2^n)}$ [$\boldsymbol{This\ is\ also\ not\ precise.}$]

$\boldsymbol{Lets\ say,}$

$\boldsymbol{T(n) = x^n}$

$\boldsymbol{Then\ again\ , we\ get}$:

$\boldsymbol{\Theta(\varphi^n)\ , where\ `\varphi\ = 1.618\ `}$

$\boldsymbol{That\ is\ Growth\ Rate\ of\ Output = \ Time\ Complexity\ here\ , but\ its\ a\ coincedence\ here\ , i.e. not\ always\ True.}$